University laboratory management in the era of digital intelligence: Computer-Aided mediation effect research on the relationship between AMO model and performance based on AMOS

Jundan Wang
Zhaoqing University
Zhaoqing, Guangdong, China
453437684@qq.com

Jiangang Feng*
School of Information Engineering
Hainan Vocational University of Science and Technology
Haikou, Hainan, China
13924119799@139.com

Abstract

With the advent of the digital intelligence age, the efficiency of laboratory management in university has received unprecedented attention. This study aims to explore how the AMO (ability, motivation, opportunity) factors of university laboratory administrators affect their job performance in the context of digital intelligence, and to analyze in depth the mediating role of reward recognition and job satisfaction in this relationship. This study used quantitative analysis methods, with 310 laboratory administrators from 10 universities in Guangdong Province, China as the research subjects. Data was collected through questionnaire surveys and statistical analysis was conducted using Amos 24.0 software. The results show that AMO has a positive impact on their work performance, reward recognition, and satisfaction; There is a significant positive correlation between satisfaction and performance; The perception of reward recognition is also significantly positively correlated with job performance. The mediating effect analysis revealed the significant parallel mediating effect of reward recognition and satisfaction between AMO and performance.This study provides empirical research support for the management of university laboratories in the digital intelligence era. It is recommended that university managers pay attention to enhancing the AMO factor of laboratory administrators and enhance their work performance by improving job satisfaction and reward recognition mechanisms.

CCS Concepts

• **General and reference** → ·Cross-computing tools and techniques; ·Empirical studies.

Keywords

Digital intelligence, Management of university laboratory, AMO model, Mediation effect, AMOS

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ICCSMT 2024, Xiamen, China
© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0999-9/24/10
https://doi.org/10.1145/3708036.3708047

ACM Reference Format:
Jundan Wang and Jiangang Feng. 2024. University laboratory management in the era of digital intelligence: Computer-Aided mediation effect research on the relationship between AMO model and performance based on AMOS. In *2024 5th International Conference on Computer Science and Management Technology (ICCSMT 2024), October 18–20, 2024, Xiamen, China.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3708036.3708047

1 Introduction

With the arrival of the digital intelligence era, the deep integration of information technology and education is driving changes in the field of higher education [1], especially in the management of university laboratories. Digital intelligence, relying on technologies such as the Internet of Things (IoT), big data analytics, artificial intelligence (AI) and machine learning, is changing the daily operation and management of laboratories. These technologies not only optimize the allocation and utilization of laboratory resources and improve the efficiency and safety of experiments, but also promote research innovation and educational quality.

Digital intelligence brings unprecedented opportunities for laboratory management. Through intelligent laboratory management systems, laboratory administrators can more effectively monitor the status of experimental equipment, track the progress of experiments in real time, as well as analyze experimental data. In addition, the digital platform facilitates interdisciplinary collaboration, promoting knowledge sharing and innovative thinking. For laboratory administrators, digital intelligence not only improves their work efficiency, but also opens up new paths for their career development.

However, digital transformation also brings a number of challenges, including data security and privacy protection issues [2], the need for training lab personnel in new technology adaptation, and the high cost of technology investment and maintenance. To meet these challenges, universities need to develop effective strategies and management measures, including strengthening cybersecurity protection, providing technology training and support, and exploring more cost-effective technology solutions.

This study uses the AMO model as a theoretical framework to explore how digital intelligence affects the abilities, motivations, and opportunities of laboratory administrators. We analyzed how digital intelligence provides laboratory administrators with new ways to improve their skills, motivate them, and provide them with more opportunities for career development. Through an online

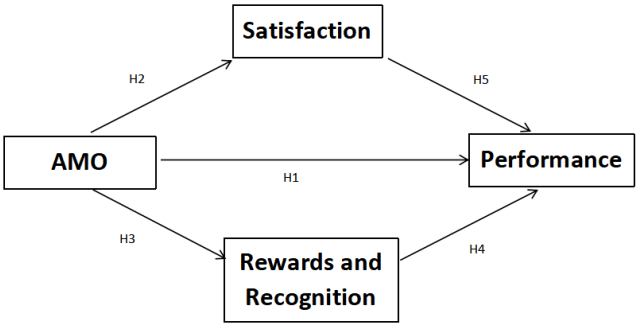

Figure 1: Hypothesized model of the study.

questionnaire, we collected data from 310 laboratory administrators from 10 universities in Guangdong Province and used AMOS software for statistical analysis to validate the relationship between the AMO model and job performance, and to explore the mediating effect of reward recognition and job satisfaction.

2 Literature review and hypothesis development

2.1 AMO theoretical model

In 2000, Appelbaum proposed the famous AMO theory through research on the effectiveness of high-performance work practices in the medical electronics, clothing, and steel industries: organizational performance is the result of the combined effects of ability, motivation, and opportunity to participate. In order to improve performance, organizations must strive to improve these three elements [3]. The AMO theory is based on a behavioral perspective and integrates the achievements of human resource theory and social exchange theory, reflecting the mechanism of high-performance work systems. Scholars have also validated the effectiveness of the AMO theory. Subramony(2009) [4] used meta-analysis to demonstrate that the combination of abilities, motivation, and opportunities has a more significant impact on organizational performance than the impact of a single practice or the entire high-performance work system.

Based on the AMO theoretical model, this study establishes the following research framework [Figure 1].

This model mainly emphasizes four key variables: AMO, performance, rewards recognition, and satisfaction. From the above model, it can be seen that managers need to strengthen the relationship between performance and reward recognition, as well as performance and satisfaction. They should combine the expected performance level of the organization with the expected reward level of employees. When employees make efforts and achieve certain results, they should actively fulfill their commitments, give appropriate rewards, and strive to balance the relationship between rewards and satisfaction.

2.2 The relationship between AMO and performance

The relationship between AMO (ability, motivation, opportunity) and performance is an important topic in the field of human resource management (HRM). We can analyze this relationship from multiple perspectives. The AMO theory suggests that employees' abilities, motivations, and opportunities are the three key factors that affect their performance. This theory has been widely applied in various studies and has been proven to play an important role in explaining employee performance [5]. For example, the AMO configuration of line managers has a positive impact on the effectiveness of implementing performance management systems, which is mediated by employees' perception of the strength of the performance management system [6]. In addition, High Performance Work Systems (HPWS) have a significant positive impact on human resource management performance in the banking industry based on the AMO model [7].

At present, a number of studies have supported the effectiveness of AMO theory in improving employee performance. For example, high-performance human resource management system can improve the performance of college students' innovation and entrepreneurship teams by influencing individual self-efficacy, internal and external motivation and organizational support [8]. In addition, the AMO-HR system has an impact on employee positive behavior through leader-member and team member exchanges, which are mediated by these exchange mechanisms [9]. Based on the literature review above, this study makes the following hypothesis:

H1: There is a significant relationship between AMO and performance.

2.3 The relationship between AMO and satisfaction

Ability is a fundamental requirement for employees to complete work tasks. When employees believe they have the skills and knowledge necessary to complete their work, they are more likely to feel satisfied [10]. In addition, ability matching, which means that an employee's abilities match their job requirements, has been proven to significantly improve internal and interpersonal satisfaction, and reduce turnover intention [11]. This indicates that improving the abilities of employees and ensuring that their abilities match job requirements is crucial for improving employee satisfaction.

Motivation is a key factor in driving employee motivation and engagement. Research has shown that motivation not only directly affects employee job satisfaction, but also indirectly affects satisfaction by influencing job performance [10, 12]. The Self Determination Theory (SDT) further elucidates how motivation affects employee behavior and attitudes, with intrinsic motivation considered an important factor in promoting employee creativity and job satisfaction [12]. This means that stimulating employees' intrinsic motivation, especially autonomous motivation, can effectively improve their job satisfaction.

Opportunity refers to the possibility for employees to participate in the decision-making process, provide suggestions, or improve work processes. Research has found that HR practices that provide opportunities to enhance employees' abilities, motivation, and engagement have a significant impact on their positive behavior, mediated by leadership member exchange and team member exchange [9]. This indicates that providing opportunities for employees to participate and contribute can enhance their satisfaction by enhancing their abilities, motivation, and sense of participation.

Based on the above literature review, this study makes the following hypothesis:

H2: There is a significant relationship between AMO and satisfaction.

2.4 The relationship between AMO and reward recognition

Reward recognition, as a means of motivation, can enhance organizational performance by enhancing employee abilities, motivation, and providing more opportunities. For example, enhancing employee abilities by providing training and development opportunities, enhancing employee motivation by setting clear goals and feedback mechanisms, and providing more opportunities by empowering employees with more autonomy and resources [13]. Therefore, this study makes the following hypothesis:

H3: There is a significant relationship between the AMO and rewards recognition.

2.5 The relationship between performance and reward recognition

Rewards recognition have a significant positive impact on employee performance. Evidence suggests that rewards recognition can enhance employee motivation, satisfaction, and motivation, thereby enhancing their performance [14–16]. In particular, recognition incentives have been found to have a very significant positive predictive effect on employee job performance [16], and the matching of reward forms and performance feedback will further improve the performance and cooperative behavior of organizational members [15]. Therefore, this study makes the following hypothesis:

H4: There is a significant relationship between rewards recognition and performance.

2.6 The relationship between performance and satisfaction

From an individual perspective, there is a positive correlation between employee job satisfaction and job performance. For example, a meta-analysis showed that the average true correlation coefficient between overall job satisfaction and job performance was 0.30 [17]. Another study also showed that there was a high degree of positive correlation between various dimensions of employee satisfaction, which had a significant positive impact on both task performance and relationship performance [18]. These studies support the view that job satisfaction has a positive effect on improving job performance.

Secondly, from an organizational perspective, there is also a positive correlation between employee satisfaction and organizational performance. A study found an expected relationship between employee satisfaction/attitude and organizational performance [19]. Another study analyzed data from 475 companies and found that employee satisfaction has a significant impact on their economic performance [20]. This indicates that improving employee satisfaction can not only enhance individual job performance, but also promote overall organizational performance improvement. Therefore, this study makes the following hypothesis:

H5: There is a significant relationship between the satisfaction and performance.

3 METHODOLOGY

This study adopts a quantitative research method and explores the mediating effect of reward recognition and job satisfaction of university laboratory administrators between AMO and performance through the AMOS structural equation model. The research design includes two main stages: questionnaire survey and data analysis.

3.1 Participants

This study adopted a random sampling method and selected laboratory administrators from 10 universities in Guangdong Province as the research subjects. A total of 326 survey questionnaires were collected, with 310 valid questionnaires and an effective rate of over 90 percent. Among these 310 laboratory administrators, 170 were male, 140 were female, In terms of education, most of the laboratory administrators are master's degree, 209 people, followed by bachelor's degree, 75 people, and the number of doctoral degrees is less, only 26 people.

3.2 Research instrument

A questionnaire was used as the research instrument for this study, which consisted of the following five sections with a total of 64 items, all of which were assessed using a 4-point Likert scale (1=strongly disagree, 2=somewhat disagree, 3=somewhat agree, and 4=strongly agree).

Basic information section: gender, age, educational background, and years of employment of the participants were collected.

AMO assessment: includes three sub-dimensions: ability, motivation and opportunity. Ability assessment consisted of 16 indicators using Carmeli and Schaubroeck's [21] scale, which was used to measure the level of professional skills and knowledge of the laboratory administrators, e.g., I have the ability to carry out standardized procedures and operational processes, good data analysis and interpretation skills, and the ability to deal with malfunctions of laboratory equipment. The internal consistency value of the ability scale in this study was 0.953.

Both motivation and opportunity assessment scales were taken from the study of Yong Xie [22]. The motivation assessment contains 12 indicators to assess work enthusiasm and work values, such as: I am very passionate about laboratory management work and have confidence and expectations for the future of my work; I can get adequate pay and rewards to motivate me, etc. The internal consistency value of the motivation scale in this study was 0.943.

The Opportunity Scale contains 8 indicators that assess the level of resources, support and opportunities received at work, e.g., I am able to receive adequate funding at work, support and assistance from my leaders, learning opportunities, etc. The internal consistency value of the opportunity scale in this study is 0.934.

Job performance assessment: includes 10 indicators and it was taken from Pedro and Juan's study [23]. It assesses the job performance, quality of task completion and development potential of laboratory managers. The internal consistency value of the job performance scale in this study was 0.915.

Reward Recognition Assessment: includes 10 indicators and it was taken from Ya Zhang's study [24]. It assesses the perceived level of rewards and recognition by laboratory administrators. The internal consistency value of the reward recognition scale in this study was 0.933.

Assessment of Job Satisfaction: includes 8 indicators and it was taken from the study of Hongbo Zhao [25]. To assess the overall satisfaction of laboratory administrators with their job and organization. The internal consistency value of job satisfaction in this study was 0.898.

In this study, the construct validity of the scales was examined using Confirmatory Factor Analysis (CFA), and the factor loadings of all the scales were greater than 0.5, indicating good construct differentiation. All model fit indices met the desirable criteria, for example, the RMSEA (root mean square error approximation) value was less than 0.08, the CFI (comparative fit index) value was close to 1.0, and the PGFI (goodness-of-fit index) and PNFI (standardized goodness-of-fit index) values were greater than 0.5, indicating that all the scales had a high level of goodness-of-fit.

3.3 Data analysis

This study used SPSS and AMOS software for data analysis, Pearson's correlation was used to determine if there is a significant relationship between AMO, performance, reward recognition, and satisfaction among laboratory administrators; and structural equation modeling structural equation modeling was used to determine the mediating effect of reward recognition, and satisfaction between AMO and performance.

In this study, Structural Equation Modeling (SEM) analysis is a powerful tool used to explore the relationship between AMO factors and job performance of university laboratory administrators. AMOS 24.0 software was selected for SEM analysis mainly due to its sophistication and user-friendliness in the field of statistical analysis. AMOS 24.0 not only supports complex model construction and estimation, but also provides rich diagnostic tools and detailed output results, which enable researchers to gain a deeper understanding of the data structure and validate hypotheses. In addition, the software's graphical interface simplifies the model specification process, making it easy to present and modify hypothesized relational paths. Compared to other SEM tools, the intuitive operation and analytical flexibility of AMOS 24.0 made it an ideal choice for this study.

4 RESULTS

This chapter presents the research findings, sequentially answers the sub questions, and provides an in-depth discussion of their relevance.

4.1 The relationship between AMO, performance, award recognition, and satisfaction of Laboratory administrator

In this study, Pearson correlation was used to statistically analyze AMO, award recognition and satisfaction of laboratory administrators. As can be seen from Table 1.

- The correlation coefficients between AMO and performance are significantly positive, and the correlation coefficients are 0.480, 0.543, and 0.470, respectively.
- The correlation coefficients between AMO and reward recognition are significantly positive, and the correlation coefficients are 0.425, 0.529, and 0.495, respectively.
- The correlation coefficients between AMO and satisfaction are significantly positive, and the correlation coefficients are 0.343, 0.419, 0.444, respectively.
- The correlation coefficient between performance and reward and recognition is significantly positive, and the correlation coefficient is 0.514.
- The correlation coefficient between performance and satisfaction is significantly positive, and the correlation coefficient is 0.398.

The above research shows that AMO model promotes the improvement of organizational performance, the improvement of reward mechanisms, and the improvement of employee satisfaction by focusing on employee abilities, motivations, and participation opportunities [26]. These factors mutually promote and influence each other, together forming a virtuous cycle of organizational development. Therefore, when implementing human resource management, university laboratories should fully consider and apply the AMO model to promote comprehensive improvement in performance, rewards, and employee satisfaction.

4.2 The mediating effect of satisfaction and reward recognition between AMO and performance

In order to test the mediating effects of satisfaction, reward and recognition on AMO and performance, a structural model of the mediating effects of satisfaction, reward and recognition on AMO and performance was constructed on Amos24.0 based on the overall sample and structural equation test was conducted [Figure 2].

Amos 24.0 was used to fit the structural equation model, and the fitting index of the structural model was compared with the ideal index. Table 2 is the model fit table. It can be seen from the table that, in terms of absolute fit index, X2/df value is 1.652, which is less than 3. The value of RMSEA is 0.046, less than 0.08. The absolute fit index fits well. From the value added fitting index, the value of IFI is 0.949, which is greater than 0.9. The value of TLI is 0.944, which is greater than 0.9. The value of CFI is 0.948, which is greater than 0.9. The value-added fitting index fits well. In terms of the simplified fitting index, the value of PGFI is 0.749, which is greater than 0.5. The value of PNFI is 0.811, which is greater than 0.5. The simplified fitting index fits well. In general, the indexes of the model fit well.

Table 3 is a path analysis table, and the path coefficient of AMO on satisfaction is significantly positive (β =0.526, P<0.001). The higher the AMO, the higher the satisfaction, proving the hypothesis H2: there is a significant relationship between AMO of laboratory administrators and satisfaction.

The path coefficient of AMO on reward recognition is significantly positive (β =0.638, P<0.001), and the higher the AMO, the

Table 1: Correlation analysis of variables.

	1	2	3	4	5	6	7
1.AMO	1						
2.Ability	.870***	1					
3.Motivation	.864***	.589***	1				
4.Opportunity	.775***	.475***	.620***	1			
5.Reward recognition	.564***	.425***	.529***	.495***	1		
6.Satisfaction	.466***	.343***	.419***	.444***	.177*	1	
7.Performance	.590***	.480***	.543***	.470***	.514***	.398***	1

Note: * * *, * *, * * respectively represent P<0.001, P<0.01, P<0.05

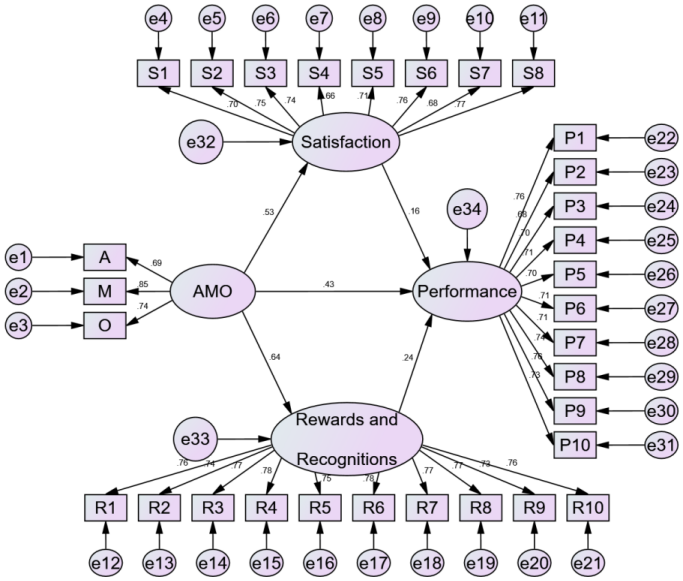

Figure 2: Structural equation model for mediating effects.

Table 2: Model fitting indicator results.

Index	Absolute Fit Index		Value Added Fitting Indicators			Simplified Fitting Index	
Classification	X2/df	RMSEA	IFI	TLI	CFI	PGFI	PNFI
Standard	<3	<0.08	>0.9	>0.9	>0.9	>0.5	>0.5
Fitting Effect	1.652	0.046	0.949	0.944	0.948	0.749	0.811

higher its reward recognition. Prove hypothesis H3: There is a significant relationship between the AMO of laboratory administrators and reward recognition.

The path coefficient of satisfaction on performance is significantly positive (β =0.160, P<0.01), and the higher the satisfaction,

Table 3: Path Analysis.

PATH			B	S.E.	C.R.	P	β
Satisfaction	<—	AMO	0.073	0.01	7.355	***	0.526
Rewards Recognition	<—	AMO	0.098	0.011	9.062	***	0.638
Performance	<—	Satisfaction	0.165	0.063	2.632	0.008	0.160
Performance	<—	AMO	0.062	0.013	4.967	***	0.431
Performance	<—	Rewards Recognition	0.229	0.065	3.528	***	0.243

Table 4: Mediation Effect Test.

EFFECT	ESTIMATE	SE	LOWER	UPPER
Total effect	0.670	0.068	0.532	0.803
Direct effect	0.431	0.147	0.156	0.727
Total indirect effects	0.239	0.093	0.066	0.437
AMO→Satisfaction→Performance	0.084	0.048	0.005	0.197
AMO→Rewards recognition→Performance	0.155	0.074	0.017	0.326

the higher the performance. Prove hypothesis H5: There is a significant relationship between the satisfaction of laboratory administrators and their performance.

The path coefficient of AMO on performance is significantly positive (β =0.431, P<0.001), and the higher the AMO, the higher its performance. Prove hypothesis H1: There is a significant relationship between the AMO of laboratory administrators and their performance.

The path coefficient of reward recognition on performance is significantly positive (β =0.243, P<0.001), and the higher the reward recognition, the higher the performance. Prove hypothesis H4: There is a significant relationship between laboratory administrator rewards recognition and performance.

Table 4 shows the test table of the mediating effect of satisfaction, reward and recognition on AMO and performance. In order to verify the mediating effect more accurately, Bootstrap method was adopted. The Bootstrap sampling times were 2000 times, the confidence interval level was set at 95%, and the sampling method was non-parametric percentile method with deviation correction. As can be seen from the table, the total effect of AMO on performance is 0.670, and its confidence interval is [0.532, 0.803], excluding 0, indicating the existence of the total effect. The direct effect is 0.431, and its confidence interval is [0.156, 0.727], excluding 0, indicating the existence of direct effect. The total indirect effect is 0.239, and its confidence interval is [0.066, 0.437], excluding 0, indicating the existence of the total indirect effect. The indirect effect of the path AMO→Satisfaction→ Performance is 0.084, and its confidence interval is [0.005, 0.197], excluding 0, indicating the existence of indirect effect, which accounts for 13 percent of the total effect. The indirect effect of the path AMO→reward and recognition→ performance is 0.155, and its confidence interval is [0.017, 0.326], excluding 0, indicating the existence of indirect effect, which accounts for 23 percent of the total effect. It can be seen that AMO can have a direct positive impact on performance, and can also have an indirect impact on performance through employee satisfaction, reward and recognition.

5 DISCUSSION

In the context of the age of digital intelligence, this study deeply explored the effects of AMO model (ability, motivation, and opportunity) on the job performance of university laboratory administrators, and analyzed the mediating effects of reward recognition and satisfaction in it. The findings indicate that AMO factors significantly and positively affect job performance, reward recognition, and job satisfaction, which is consistent with theoretical expectations. However, in order to more fully understand the deeper mechanisms of these relationships in the context of digital intelligence, this study further analyzes how digital intelligence enhances or limits the role of each AMO dimension.

5.1 The Impact of Digital Intelligence on AMO

Digital intelligence brings many positive impacts to university laboratory administrators. It provides advanced technological tools and platforms to enhance managerial abilities, digital resources to facilitate access to information and enhance professional skills, and data analytics and decision-support systems to improve problem-solving abilities and efficiency. Digital Intelligence also provides real-time feedback and incentives to enhance employee motivation, and digital performance management systems to capture and reward performance in a timely manner, promoting work input and performance improvement. At the same time, digital intelligence provides more opportunities for participation in decision-making and access to resources, and the collaborative and sharing nature of digital platforms enhances employees' sense of engagement and belonging.

However, digital intelligence may also limit the role of AMO factors. Rapid technological change may challenge the ability of some employees to keep up. Over-reliance on digital tools may weaken

interpersonal and communication skills, affecting teamwork and work atmosphere.

Based on this, it is recommended that university laboratory managers make full use of digital intelligence tools to enhance employee abilities, motivation, and engagement opportunities, while at the same time paying attention to possible problems that may arise from the use of technology, such as over-reliance, skill degradation, and burnout, and taking measures to ensure that digital intelligence effectively contributes to employee job performance and satisfaction.

5.2 New ways to improve performance

This study provides an in-depth analysis of the relationship between AMO, performance, reward recognition, and satisfaction of university laboratory administrators. Pearson's correlation analysis and AMOS structural equation modeling revealed a significant positive correlation between these variables, with reward recognition and satisfaction mediating the juxtaposed effects between AMO and performance.

Ability, motivation, and opportunity interact with each other to promote efficient performance among laboratory administrators. For example, ability provides a knowledge and skill base, motivation stimulates work, and opportunity is a platform for the expression of talent. Satisfaction is positively related to performance, with satisfied employees investing more energy and enthusiasm and working at a higher quality, and is also associated with low turnover and high team cohesion. Reward recognition is positively related to performance. Being recognized and rewarded for work satisfies intrinsic motivation, increases satisfaction and loyalty, and stimulates motivation and creativity. Structural equation modeling reveals the juxtaposed mediating effects of reward recognition and satisfaction, providing new perspectives for understanding the mechanisms influencing performance.

Based on this, recommendations are made to enhance performance: strengthen ability development and training, and provide opportunities for continuous career development and skills training; stimulate intrinsic motivation, set challenging goals and provide space for autonomous decision-making, and implement personalized incentives; create favourable work opportunities, and participate in important projects and academic exchanges; set up a fair reward and recognition mechanism to ensure that outstanding performance is rewarded in a timely and just manner; enhance job satisfaction, and Provide a favorable working environment and opportunities to participate in decision-making; adopt comprehensive human resource management strategies and establish a feedback mechanism. Through these strategies, the performance of laboratory administrators can be improved, and the quality of research and teaching can be enhanced.

6 CONCLUSION

This study reveals the positive impact of AMO model on the job performance of university laboratory managers in the context of digital intelligence and the side-by-side mediating effects of reward recognition and job satisfaction in it. The findings highlight the positive effects of digital intelligence in enhancing employee ability, motivation, and opportunities, while also pointing out the challenges it may pose. These findings provide valuable insights for university laboratory administration to make full use of digital tools and resources to optimize employee ability development, motivation, and engagement opportunities to promote employee performance and satisfaction during the digital intelligence transformation process. Future research could further explore the dynamics of AMO models in the context of digital intelligence and how different organizational environments and cultures affect the relationship between AMO and job performance.

References

[1] He Kekang. How to achieve the "deep integration" of information technology and education [J]. Curriculum, Textbooks, Teaching Methods, 2014, 34 (02): 58-62+67.

[2] H. Stewart. "Digital Transformation Security Challenges." Journal of Computational Information Systems. 2022, 919 - 936.

[3] Appelbaum,E.,Bailey,T. R. ,Berg, P.B., & Kalleberg, A.L. 2000. Manufacturing advantage:Why high-performance work systems pay off. Academy of Management Review, 26(3), 459-462.

[4] Mahesh Subramony. 2009. A meta-analytic investigation of the relationship between HRM bundles and firm performance. , 48(5), 745–768.

[5] Ashlea Kellner, Kenneth Cafferkey, Keith Townsend. Elgar Introduction to Theories of Human Resources and Employment Relations[M]. 2019. DOI:https://doi.org/10.4337/9781786439017.

[6] Thomas Van Waeyenberg, Adelien Decramer. Line managers'AMO to manage employees' performance: the route to effective and satisfying performance management[J].The International Journal of Human Resource Management. 2018 (29):3093-3114.

[7] Fatimah Mohamed Mahdy, and Asia Yagoub AIhadi.The effect of high-performance work systems according to AMO model on HRM performance outcomes: An analytical study on the banking sector[J].Management Science Letters. 2021, 1755–1764.

[8] Qin Beibei, Xiong Yan, Li Ruijie. The Impact of High Performance Human Resource Management on the Performance of College Students' Innovation and Entrepreneurship Teams: Based on AMO Theory [J] Chinese and Foreign Entrepreneurs. 2020 (17): 121-123.

[9] Ahmad Adnan Al-Tit.The impact of AMO-HR systems on proactive employee behavior:The mediating contribution of leader-member and team-member exchange[J]. International Journal of Engineering Business management. 2020. https://doi.org/10.1177/184797902094723.

[10] Jasiyah, R. 2018. The effect of ability and motivation on job satisfaction and employee performance. Archives of Business Research, 6(12). https://doi.org/10.14738/abr.612.5644.

[11] Cheng Junjun, Su Yong, Wang Yuyang. Ability matching, employability, and turnover intention: the mediating role of job satisfaction [J]. Contemporary Finance and Economics. 2015 (02): 66-75.

[12] Daniel Patrik Biskupski. The Impact of Motivation on Employee Job Satisfaction, Performance, and Creativity [D]. 2020. Southwest University of Finance and Economics.

[13] Zeyad Almutawa , Nuttawuth Muenjohn , Jiaying Zhang.The effect of human resource management system on employees' commitment: The mediating role of the AMO model[J]. The Journal of Developing Areas. 2016 50(06):17-29.

[14] Vera Akafo, P. Boateng. Impact of Reward and Recognition on Job Satisfaction and Motivation[J]. 2015. Business, Education.European Journal of Business and Management.

[15] Dai Wenting, Shi Kan, Han Xiaoyan, Zhou Xinyue. The impact of reward value orientation on performance feedback [J]. Psychological Science. 2013, 36 (06): 1451-1458.

[16] Fan Wenyu, Song Hongfeng. Research on the impact of recognition incentives on employee performance [J]. China Forestry Economy. 2019 (03): 63-65+74.

[17] Judge, T. A., Thoresen, C. J., Bono, J. E., & Patton, G. K. 2001. The job satisfaction–job performance relationship: A qualitative and quantitative review. Psychological Bulletin, 127(3):376-407. https://doi.org/10.1037/0033-2909.127.3.376.

[18] Xu Qin. Empirical Analysis of the Impact of Knowledge Employee Satisfaction on Work Performance [J]. Statistics and Decision Making. 2015 (05):117-119.

[19] Ostroff, C. 1992. The relationship between satisfaction, attitudes, and performance: An organizational level analysis. Journal of Applied Psychology, 77(6), 963–974. https://doi.org/10.1037/0021-9010.77.6.963.

[20] Melián-González, Santiago; Bulchand-Gidumal, Jacques; González López-Valcárcel, Beatriz 2015, New evidence of the relationship between employee satisfaction and firm economic performance. 44(6),906 - 929.

[21] Carmeli A, Schaubroeck J. The influence of leaders' and other referents' normative expectations on individual involvement in creative work[J]. The Leadership Quarterly. 2006 (18),35-48.DOI:10.1016/j.leaqua.2006.11.001

[22] Xie Yong. Diagnosis and improvement suggestions for human resource management in FL company based on the AMO model[D]. University of Electronic Science and Technology of China, 2018.

[23] Ramos Villagrasa Pedro J.;Barrada Juan R.;Fernández del Río Elena;Koopmans Linda.Assessing Job Performance Using Brief Self-report Scales: The Case of the Individual Work Performance Questionnaire[J]. Revista de Psicología del Trabajoy de las Organizaciones. 2019, (35):195-205.https://doi.org/10.5093/jwop2019a21.

[24] Zhang Ya. Research on Employee Innovation Performance Based on AMO Theory: The Mediating Role of Improvisation Behavior [D]. Northern Polytechnical University, 2020.

[25] Zhao Hongbo. Research on Satisfaction Evaluation and Optimization of Incentive Work for Civil Servants in the Grassroots Tax System of City C[D]. Northwestern University, 2021.

[26] Àngels Dasí , Torben Pedersen, Lívia Lopes Barakat, Tiago Rangel Alves.Teams and Project Performance: An Ability, Motivation, and Opportunity Approach[J]. Project Management Journal. 2021, 2:1-15.